Query | Query History

```
--1 Create an index on the actual_departure column in the flights table.
CREATE INDEX index_actual_departure
ON flights (actual_departure);

--2 Create a unique index to ensure flight_no and scheduled_departure combinations are unique.
CREATE UNIQUE INDEX index_flight_no_scheduled_departure
ON flights (flight_no,scheduled_departure);
--не раннится
SELECT flight_no,scheduled_departure,
COUNT(*) AS duplicate_count
FROM flights
GROUP BY flight_no, scheduled_departure
HAVING COUNT(*) > 1;

--3 Create a composite index on the departure_airport_id and arrival_airport_id columns.
CREATE INDEX index_departure_airport_id_arrival_airport_id
ON flights (departure_airport_id,arrival_airport_id);

--4 Evaluate the difference in query performance with and without indexes. Measure performance di
DROP INDEX index_departure_airport_id_arrival_airport_id;
```

Data Output | Messages | Notifications

```
CREATE INDEX

Query returned successfully in 79 msec.
```

```
--2 Create a unique index to ensure flight_no and scheduled_departure combinations are unique.
CREATE UNIQUE INDEX index_flight_no_scheduled_departure
ON flights (flight_no,scheduled_departure);
--не раннится
SELECT flight_no,scheduled_departure,
COUNT(*) AS duplicate_count
FROM flights
GROUP BY flight_no, scheduled_departure
HAVING COUNT(*) > 1;

--3 Create a composite index on the departure_airport_id and arrival_airport_id columns.
CREATE INDEX index_departure_airport_id_arrival_airport_id
ON flights (departure_airport_id,arrival_airport_id);

--4 Evaluate the difference in query performance with and without indexes. Measure performance diffe
DROP INDEX index_departure_airport_id_arrival_airport_id;
```

Data Output | Messages | Notifications

```
CREATE INDEX

Query returned successfully in 65 msec.
```

```sql
--2 Create a unique index to ensure flight_no and scheduled_departure combinations are unique.
CREATE UNIQUE INDEX index_flight_no_scheduled_departure
ON flights (flight_no,scheduled_departure);
--не раннится
SELECT flight_no,scheduled_departure,
COUNT(*) AS duplicate_count
FROM flights
GROUP BY flight_no, scheduled_departure
HAVING COUNT(*) > 1;

--3 Create a composite index on the departure_airport_id and arrival_airport_id columns.
CREATE INDEX index_departure_airport_id_arrival_airport_id
ON flights (departure_airport_id,arrival_airport_id);

--4 Evaluate the difference in query performance with and without indexes. Measure performance dif
DROP INDEX index_departure_airport_id_arrival_airport_id;
```

Data Output | Messages | Notifications

SQL

	flight_no character varying (50)	scheduled_departure date	duplicate_count bigint

```sql
--3 Create a composite index on the departure_airport_id and arrival_airport_id columns.
CREATE INDEX index_departure_airport_id_arrival_airport_id
ON flights (departure_airport_id,arrival_airport_id);

--4 Evaluate the difference in query performance with and without indexes. Measure performance differe
DROP INDEX index_departure_airport_id_arrival_airport_id;

EXPLAIN ANALYZE
SELECT *
FROM flights
WHERE departure_airport_id = 1
AND arrival_airport_id = 5;

CREATE INDEX index_departure_airport_id_arrival_airport_id
ON flights (departure_airport_id, arrival_airport_id);

EXPLAIN ANALYZE
```

Data Output | Messages | Notifications

```
CREATE INDEX

Query returned successfully in 78 msec.
```

Query | Query History

```sql
--4 Evaluate the difference in query performance with and without indexes. Measure performance differences.
DROP INDEX index_departure_airport_id_arrival_airport_id;

EXPLAIN ANALYZE
SELECT *
FROM flights
WHERE departure_airport_id = 1
AND arrival_airport_id = 5;

CREATE INDEX index_departure_airport_id_arrival_airport_id
ON flights (departure_airport_id, arrival_airport_id);

EXPLAIN ANALYZE
```

Data Output | Messages | Notifications

Showing rows: 1 to 12 | Page No: 1 of 1

	QUERY PLAN text
1	Bitmap Heap Scan on flights (cost=4.30..9.97 rows=2 width=61) (actual time=0.414..0.416 rows=1.00 loops=1)
2	Recheck Cond: ((departure_airport_id = 1) AND (arrival_airport_id = 5))
3	Heap Blocks: exact=1
4	Buffers: shared hit=1 read=2
5	-> Bitmap Index Scan on index_departure_airport_id_arrival_airport_id (cost=0.00..4.29 rows=2 width=0) (actual time=0.382..0.383 rows=1.00 loo...
6	Index Cond: ((departure_airport_id = 1) AND (arrival_airport_id = 5))
7	Index Searches: 1
8	Buffers: shared read=2

```
431  --5 Use EXPLAIN ANALYZE to check index usage in a query filtering by departure_airport and arrival_airport.
432  EXPLAIN ANALYZE
433  SELECT *
434  FROM flights
435  WHERE departure_airport_id = 1
436  AND arrival_airport_id = 5;
437
438  --6 Create a unique index for the passport_number of the Passengers table. Check if the index was created or not
439  CREATE UNIQUE INDEX index_passengers_passport_number
440  ON passengers (passport_number);
441
442  SELECT indexname, indexdef
443  FROM pg_indexes
```

Data Output | Messages | Notifications

Showing rows: 1 to 10 | Page No: 1 of 1

	QUERY PLAN text
1	Bitmap Heap Scan on flights (cost=4.30..9.97 rows=2 width=61) (actual time=1.227..1.229 rows=1.00 loops=1)
2	Recheck Cond: ((departure_airport_id = 1) AND (arrival_airport_id = 5))
3	Heap Blocks: exact=1
4	Buffers: shared hit=3
5	-> Bitmap Index Scan on index_departure_airport_id_arrival_airport_id (cost=0.00..4.29 rows=2 width=0) (actual time=1.117..1.118 rows=1.00 loo…
6	Index Cond: ((departure_airport_id = 1) AND (arrival_airport_id = 5))
7	Index Searches: 1
8	Buffers: shared hit=2
9	Planning Time: 12.061 ms
10	Execution Time: 2.060 ms

Query | Query History

```
438  --6 Create a unique index for the passport_number of the Passengers table. Check if the index was created or
439  CREATE UNIQUE INDEX index_passengers_passport_number
440  ON passengers (passport_number);
441
442  SELECT indexname, indexdef
443  FROM pg_indexes
444  WHERE tablename = 'passengers';
445
446  SELECT column_name, data_type, is_nullable
447  FROM information_schema.columns
448  WHERE table_name = 'passengers'
449  ORDER BY ordinal_position;
450
```

Data Output | Messages | Notifications

Showing rows: 1 to 4 | Page No: 1 of 1

	indexname name	indexdef text
1	passengers_pkey	CREATE UNIQUE INDEX passengers_pkey ON public.passengers USING btree (passenger_id)
2	unique_passport_number	CREATE UNIQUE INDEX unique_passport_number ON public.passengers USING btree (passport_number)
3	idx_passengers_lower_last_name	CREATE INDEX idx_passengers_lower_last_name ON public.passengers USING btree (lower((last_name)::text))
4	index_passengers_passport_num…	CREATE UNIQUE INDEX index_passengers_passport_number ON public.passengers USING btree (passport_nu…

```
438  --6 Create a unique index for the passport_number of the P
439  CREATE UNIQUE INDEX index_passengers_passport_number
440  ON passengers (passport_number);
441
442  SELECT indexname, indexdef
443  FROM pg_indexes
444  WHERE tablename = 'passengers';
445
446  SELECT column_name, data_type, is_nullable
447  FROM information_schema.columns
448  WHERE table_name = 'passengers'
449  ORDER BY ordinal_position;
450
```

Data Output | Messages | Notifications

	column_name name	data_type character varying	is_nullable character varying (3)
1	passenger_id	integer	NO
2	first_name	character varying	YES
3	last_name	character varying	YES
4	date_of_birth	date	YES
5	gender	character varying	YES
6	country_of_citizens...	character varying	YES
7	country_of_residence	character varying	YES
8	passport_number	character varying	YES
9	created_at	date	YES
10	update_at	date	YES

```sql
--7 Create an index for the Passengers table. Use for that first name, last name, date of birth and country of c
CREATE INDEX index_passengers_details
ON passengers (first_name, last_name, date_of_birth, country_of_citizenship);

SELECT *
FROM passengers
WHERE country_of_citizenship = 'Philippines'
AND date_of_birth >= '1984-01-01'
AND date_of_birth < '1985-01-01';

EXPLAIN ANALYZE
SELECT *
FROM passengers
WHERE country_of_citizenship = 'Philippines'
AND date_of_birth >= '1984-01-01'
AND date_of_birth < '1985-01-01';

--8 Write a SQL query to list indexes for table Passengers. After delete the created indexes.
SELECT indexname, indexdef
```

Data Output | Messages | Notifications

Showing rows: 1 to 6 | Page No: 1 of 1

	QUERY PLAN text
1	Seq Scan on passengers (cost=0.00..6.54 rows=1 width=62) (actual time=0.016..0.035 rows=1.00 loops=1)
2	Filter: ((date_of_birth >= '1984-01-01'::date) AND (date_of_birth < '1985-01-01'::date) AND ((country_of_citizenship)::text = 'Philippines':...
3	Rows Removed by Filter: 201
4	Buffers: shared hit=3
5	Planning Time: 0.104 ms
6	Execution Time: 0.049 ms

```sql
--8 Write a SQL query to list indexes for table Passengers. After delete the created indexes.
SELECT indexname, indexdef
FROM pg_indexes
WHERE tablename = 'passengers';

DROP INDEX IF EXISTS idx_passengers_passport_number;
DROP INDEX IF EXISTS idx_passengers_name_dob_couDntry;

SELECT indexname, indexdef
FROM pg_indexes
WHERE tablename = 'passengers';

--9 Create an index on the passenger_id column of the booking table. Using EXPLAIN ANALYZE, compare the execution pla
EXPLAIN ANALYZE
SELECT *
FROM booking
WHERE passenger_id = 100;

CREATE INDEX idx_booking_passenger_id
```

Data Output | Messages | Notifications

Showing rows: 1 to 5 | Page No: 1 of 1

	indexname name	indexdef text
	passengers_pkey	CREATE UNIQUE INDEX passengers_pkey ON public.passengers USING btree (passenger_id)
2	unique_passport_number	CREATE UNIQUE INDEX unique_passport_number ON public.passengers USING btree (passport_number)
3	idx_passengers_lower_last_name	CREATE INDEX idx_passengers_lower_last_name ON public.passengers USING btree (lower((last_name)::text))
4	index_passengers_passport_num...	CREATE UNIQUE INDEX index_passengers_passport_number ON public.passengers USING btree (passport_number)
5	index_passengers_details	CREATE INDEX index_passengers_details ON public.passengers USING btree (first_name, last_name, date_of_birth, country_of_citize...

```sql
--9 Create an index on the passenger_id column of the booking table. Using EXPLAIN ANALYZE, c
EXPLAIN ANALYZE
SELECT *
FROM booking
WHERE passenger_id = 100;

CREATE INDEX idx_booking_passenger_id
ON booking(passenger_id);

EXPLAIN ANALYZE
SELECT *
FROM booking
WHERE passenger_id = 100;

```

Data Output | Messages | Notifications

Showing rows: 1 to 10 | Page No:

	QUERY PLAN text
1	Bitmap Heap Scan on booking (cost=4.17..11.51 rows=3 width=50) (actual time=0.084..0.089 rows=3.00 loops=1)
2	Recheck Cond: (passenger_id = 100)
3	Heap Blocks: exact=2
4	Buffers: shared hit=3
5	-> Bitmap Index Scan on idx_booking_passenger_id (cost=0.00..4.17 rows=3 width=0) (actual time=0.033..0.034 rows=3.00 loo...
6	Index Cond: (passenger_id = 100)
7	Index Searches: 1
8	Buffers: shared hit=1
9	Planning Time: 0.182 ms
10	Execution Time: 0.134 ms

Query | Query History

```
494  --10 Create a composite index on the booking_flight table using the booking_id and flight_id
495  CREATE INDEX idx_booking_flight_composite
496  ON booking_flight(booking_id, flight_id);
497
498  SELECT f.flight_no,b.price
499  FROM booking b
500  JOIN booking_flight bf
501  ON b.booking_id = bf.booking_id
502  JOIN flights f
503  ON bf.flight_id = f.flight_id
504  WHERE b.price > 9000;
505
506  EXPLAIN ANALYZE
507  SELECT f.flight_no,b.price
508  FROM booking b
509  JOIN booking_flight bf
510  ON b.booking_id = bf.booking_id
511  JOIN flights f
```

Data Output | Messages | Notifications

Showing rows: 1 to 24 | Page No

	QUERY PLAN text
1	Hash Join (cost=39.22..67.18 rows=121 width=12) (actual time=0.531..0.903 rows=118.00 loops=1)
2	Hash Cond: (f.flight_id = bf.flight_id)
3	Buffers: shared hit=31
4	-> Seq Scan on flights f (cost=0.00..23.00 rows=1000 width=10) (actual time=0.020..0.165 rows=1000.00 loops=1)
5	Buffers: shared hit=13
6	-> Hash (cost=37.71..37.71 rows=121 width=10) (actual time=0.504..0.507 rows=118.00 loops=1)
7	Buckets: 1024 Batches: 1 Memory Usage: 14kB
8	Buffers: shared hit=18

Query | Query History

```
515  --11 Create an index on the status column of the flights table. Run EXPLAIN ANALYZE f
516  CREATE INDEX idx_flights_status
517  ON flights(status);
518
519  EXPLAIN ANALYZE
520  SELECT *
521  FROM flights
522  WHERE status = 'Departed';
523
524  EXPLAIN ANALYZE
525  SELECT *
526  FROM flights
527  WHERE status = 'Boarding';
528
529  --12 Create an index on the weight_in_kg column of the baggage table. Compare the pla
530  CREATE INDEX idx_flights_status
531  ON flights(status);
532
```

Data Output | Messages | Notifications

Showing rows: 1 to 10

	QUERY PLAN text
1	Bitmap Heap Scan on flights (cost=4.19..14.95 rows=5 width=61) (actual time=0.673..0.675 rows=5.00 loops=1)
2	Recheck Cond: ((status)::text = 'Departed'::text)
3	Heap Blocks: exact=1
4	Buffers: shared hit=2
5	-> Bitmap Index Scan on idx_flights_status (cost=0.00..4.19 rows=5 width=0) (actual time=0.619..0.619 rows=5.00 loo…
6	Index Cond: ((status)::text = 'Departed'::text)
7	Index Searches: 1
8	Buffers: shared hit=1

```
522 WHERE status = 'Departed';
523
524 EXPLAIN ANALYZE
525 SELECT *
526 FROM flights
527 WHERE status = 'Boarding';
528
529 --12 Create an index on the weight_in_kg column of the baggage table. Compare the
530 CREATE INDEX idx_flights_status
531 ON flights(status);
532
```

Data Output | Messages | Notifications

Showing rows: 1 to 6

	QUERY PLAN text
1	Seq Scan on flights (cost=0.00..25.50 rows=500 width=61) (actual time=0.164..0.383 rows=500.00 loop…
2	Filter: ((status)::text = 'Boarding'::text)
3	Rows Removed by Filter: 500
4	Buffers: shared hit=13
5	Planning Time: 9.815 ms
6	Execution Time: 1.265 ms

```sql
--12 Create an index on the weight_in_kg column of the baggage table. Compare the
CREATE INDEX idx_baggage_weight ON baggage (weight_in_kg);

-- Аз бөлігі → Index Scan
EXPLAIN ANALYZE SELECT * FROM baggage WHERE weight_in_kg > 49;

-- Барлығы дерлік → Seq Scan
EXPLAIN ANALYZE SELECT * FROM baggage WHERE weight_in_kg > 5;

--13 Create an index on the airport_name column of the airport table. Check with
CREATE INDEX idx_airport_name
ON airport(airport_name);

EXPLAIN ANALYZE
SELECT *
FROM airport
WHERE airport_name LIKE 'A%';
```

Data Output | Messages | Notifications

Showing rows: 1 to 6

	QUERY PLAN text
1	Seq Scan on baggage (cost=0.00..11.25 rows=451 width=21) (actual time=0.033..0.453 rows=451.00 loop…
2	Filter: (weight_in_kg > '5'::numeric)
3	Rows Removed by Filter: 49
4	Buffers: shared hit=5
5	Planning Time: 0.061 ms
6	Execution Time: 0.483 ms

```
538  --13 Create an index on the airport_name column of the airport table. Check with
539  CREATE INDEX idx_airport_name
540  ON airport(airport_name);
541
542  EXPLAIN ANALYZE
543  SELECT *
544  FROM airport
545  WHERE airport_name LIKE 'A%';
546
547  EXPLAIN ANALYZE
548  SELECT *
549  FROM airport
550  WHERE airport_name LIKE '%Airport';
551
552  --14 Create a composite index on the country and city columns of the airport tabl
553  CREATE INDEX idx_airport_country_city
```

Data Output | Messages | Notifications

Showing rows: 1 to 6

	QUERY PLAN text
1	Seq Scan on airport (cost=0.00..1.26 rows=4 width=59) (actual time=0.008..0.012 rows=18.00 loop…
2	Filter: ((airport_name)::text ~~ '%Airport'::text)
3	Rows Removed by Filter: 3
4	Buffers: shared hit=1
5	Planning Time: 0.063 ms
6	Execution Time: 0.021 ms

Query | Query History

```
552  --14 Create a composite index on the country and city columns of the airport table.
553  CREATE INDEX idx_airport_country_city
554  ON airport(country, city);
555
556  EXPLAIN ANALYZE
557  SELECT *
558  FROM airport
559  WHERE country = 'Kazakhstan';
560
561  EXPLAIN ANALYZE
562  SELECT *
563  FROM airport
564  WHERE city = 'Almaty';
565
566  EXPLAIN ANALYZE
567  SELECT *
568  FROM airport
569  WHERE country = 'Kazakhstan'
570  AND city = 'Almaty';
```

Data Output | Messages | Notifications

Showing rows: 1 to 6

	QUERY PLAN text
1	Seq Scan on airport (cost=0.00..1.31 rows=1 width=59) (actual time=0.017..0.017 rows=1.00 loop…
2	Filter: (((country)::text = 'Kazakhstan'::text) AND ((city)::text = 'Almaty'::text))
3	Rows Removed by Filter: 20
4	Buffers: shared hit=1
5	Planning Time: 0.258 ms
6	Execution Time: 0.024 ms

```
572  --15 Create an expression (functional) index on the lower-case last_name of the Passen
573  CREATE INDEX idx_passengers_lower_last_name
574  ON passengers(LOWER(last_name));
575
576  EXPLAIN ANALYZE
577  SELECT *
578  FROM passengers
579  WHERE LOWER(last_name) = 'irnis';
580
581  EXPLAIN ANALYZE
582  SELECT *
583  FROM passengers
584  WHERE last_name = 'Irnis';
585
586  --16 Create a covering index on the booking table for the columns passenger_id and pri
587  CREATE INDEX idx_booking_passenger_price
588  ON booking(passenger_id, price);
589
```

Data Output | Messages | Notifications

Showing rows: 1 to 6

	QUERY PLAN text
1	Seq Scan on passengers (cost=0.00..5.53 rows=1 width=62) (actual time=0.015..0.034 rows=1.00 loop…
2	Filter: ((last_name)::text = 'Irnis'::text)
3	Rows Removed by Filter: 201
4	Buffers: shared hit=3
5	Planning Time: 0.092 ms
6	Execution Time: 0.047 ms

```sql
--16 Create a covering index on the booking table for the columns passenger_id and price. Wri
CREATE INDEX idx_booking_passenger_price
ON booking(passenger_id, price);

EXPLAIN ANALYZE
SELECT price
FROM booking
WHERE passenger_id = 100;

--17 Try to create a unique index on the airline_code column of the airline table. If an erro
CREATE UNIQUE INDEX idx_airline_code_unique
ON airline(airline_code);

SELECT airline_code, COUNT(*)
```

Data Output | Messages | Notifications

Showing rows: 1 to 10 | Page No:

	QUERY PLAN text
1	Bitmap Heap Scan on booking (cost=4.17..11.51 rows=3 width=6) (actual time=1.695..1.704 rows=3.00 loops=1)
2	Recheck Cond: (passenger_id = 100)
3	Heap Blocks: exact=2
4	Buffers: shared hit=3
5	-> Bitmap Index Scan on idx_booking_passenger_id (cost=0.00..4.17 rows=3 width=0) (actual time=1.244..1.244 rows=3.00 loo…
6	Index Cond: (passenger_id = 100)
7	Index Searches: 1
8	Buffers: shared hit=1
9	Planning Time: 17.430 ms
10	Execution Time: 2.871 ms

```sql
--17 Try to create a unique index on the airline_code column c
CREATE UNIQUE INDEX idx_airline_code_unique
ON airline(airline_code);

SELECT airline_code, COUNT(*)
FROM airline
GROUP BY airline_code
HAVING COUNT(*) > 1;

SELECT *
FROM airline
WHERE airline_code IN (
SELECT airline_code
FROM airline
GROUP BY airline_code
HAVING COUNT(*) > 1)
ORDER BY airline_code;

CREATE UNIQUE INDEX idx_airline_code_unique
ON airline(airline_code);
```

ata Output | Messages | Notifications

```
REATE INDEX

uery returned successfully in 44 msec.
```